Computational Phase Synchronization

Causal Alignment, Validation-Gated Fusion, and Failure Conditions

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Abstract

Distributed systems often receive several partial measurements of the same changing process, but those measurements may not arrive in one common cyclic frame. A camera, microphone, inertial sensor, memory module, or artificial agent can preserve useful content while its timing or phase relation differs from that of its peers. Averaging such states before estimating their relative offsets can cancel signal, mix incompatible observations, or create a confident but incorrect fused state.

Existing engineering provides several parts of the solution. Hardware synchronization aligns clocks before measurement. Attention and scalar gating weight inputs without necessarily retaining circular relations. Angular synchronization estimates unknown angles from noisy pairwise offsets. Multiscale aperture synthesis imaging demonstrates that distributed optical measurements can retain complex wavefield information, undergo computational relative-phase optimization, and then be coherently fused. Recent artificial systems also use complex activations, phase relationships, recurrent constraint satisfaction, coupled oscillators, or neural synchronization as internal representations.

This paper isolates one narrower operation. Each module retains a complex-valued state; a causal estimator compares each module with a reference, tracks the relative offset, rotates states into a common gauge, and fuses them only when held-out validation shows a task benefit. The operation is called **computational phase synchronization** after the problem and mechanism have been specified.

An executable synthetic benchmark tests five declared worlds over 48 seeds each. Adaptive alignment was accepted in every stable-offset and slow-drift-with-jump run. Median normalized mean-squared error fell from 0.014789 to 0.001865 in the stable world and from 0.014854 to 0.002591 in the slow-drift world. The same estimator failed under fast drift, unpredictable offsets, and distractor modules. A validation gate rejected it in every run of those three worlds and reverted to the reference-module baseline. The result therefore supports a conditional engineering claim: explicit phase alignment can help when modules observe a shared state and relative offsets are predictable at the estimator's bandwidth; validation-gated abstention is necessary when those conditions fail.

A preregistered semi-synthetic extension then replaced the generated latent trajectory with the analytic signal of a checksum-verified five-minute human ECG record while retaining controlled module offsets, noise, and distractors. Across 48 perturbation seeds per world, selected median NMSE fell from 0.008947 to 0.001173 under stable offsets, from 0.008956 to 0.001327 under slow drift with a jump, and from 0.008955 to 0.002356 in the declared fast-drift condition. The gate rejected every unpredictable-offset and distractor run and returned the reference unchanged.

Together, the benchmarks validate estimator and failure-test logic in declared synthetic and semi-synthetic data-generating processes. They do not establish a general AI-task advantage, a clinical result, a biological neural-phase mechanism, Neural Array Projection Oscillation Tomography, Phase Wave Differentials, Self Aware Networks, or consciousness.

Scope and evidence boundary

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1. The Problem Before the Term

Imagine eight instruments watching one moving process. Each instrument reports a useful local state, but seven are rotated around a circle relative to the first. If their magnitudes are averaged while those rotations remain unknown, the instruments can cancel one another even when every instrument is accurate. If the rotations drift, a correction learned yesterday can become wrong today. If some instruments are watching a different process, forcing them into one frame can manufacture apparent agreement.

The computational problem has four parts:

1. preserve the relation that was lost by premature scalar averaging;
2. estimate only the relative offsets that the observations identify;
3. update those estimates causally when offsets drift; and
4. refuse fusion when held-out evidence says alignment is not helping.

The fourth part matters as much as the first three. A system that always declares synchrony will turn unrelated signals into an apparently coherent state. The proposed architecture therefore treats abstention as a first-class output rather than as a post hoc disclaimer.

2. What Established Methods Already Supply

2.1 Hardware synchronization

Shared clocks, triggers, and calibrated acquisition paths can align sensors before data are recorded. This remains preferable when it is feasible. The present problem arises when pre-alignment is unavailable, imperfect, expensive, or insufficiently stable.

2.2 Scalar gates and attention

Learned weights can select or suppress modules. Attention can create content-dependent mixtures and communicate information across tokens or modalities [6]. The inherited Draft 6 comparator family includes weighted averaging, learned scalar gates, and mixture-of-experts routing. These methods can indirectly learn timing relations, but an ordinary scalar weight does not explicitly represent a circular offset, and an attention score does not by itself solve a gauge-identification problem.

2.3 Angular synchronization

Angular synchronization is an established inverse problem. It estimates unknown angles from noisy relative-angle observations and recovers them only up to a global additive phase. Singer's eigenvector and semidefinite formulations make the gauge structure and robustness problem explicit [2]; later analysis gives near-optimal bounds for this established problem [8]. The estimator below is not presented as the invention of angular synchronization. It is a small causal reference-based baseline chosen to make the proposed module-fusion test executable and falsifiable.

2.4 Complex-valued and synchronization-based neural systems

Complex-valued neural networks long predate this paper [7]. SynCx uses complex-valued recurrent computation and phase relations for iterative object binding [9]. Continuous Thought Machines use neural synchronization as a latent representation [10]. BRICK represents graph nodes as coupled oscillators and reports task gains from synchronization-driven dynamics [11]. These systems establish that phase-like variables in AI are a populated field, not an empty novelty claim.

The distinct question here is narrower: can a causal relative-offset estimator be paired with an explicit held-out acceptance gate so that it improves fusion under recoverable drift and abstains under declared failure worlds?

3. From Plain Operation to Named Architecture

The operation can be understood without a new term:

1. Each local module emits both magnitude and circular position.
2. One module supplies a reference gauge.
3. Pairwise products reveal how the other modules are rotated relative to that reference.
4. A running circular estimate tracks each relative rotation.
5. Each local state is counter-rotated into the reference frame.
6. The aligned states are fused.
7. Validation data determine whether the adaptive fusion is allowed to replace a safer reference baseline.

Only after those steps are clear is the operation named **computational phase synchronization**. The name refers to computational estimation and alignment of relative cyclic states; it does not claim that every input is a physical wave or that all useful representations must synchronize.

The causal chain is:

```
local state
-> preserve circular relation
-> estimate relative offset from past observations
-> rotate into a common gauge
-> test held-out task error
-> fuse or abstain
```

4. Formal Model and Identifiability

Let the shared latent state at discrete time t be

$$s_t = a_t e^{i\psi_t}, \qquad a_t > 0. \tag{1}$$

Module m observes that state through a module-specific offset $\theta_{m,t}$ and complex noise $\epsilon_{m,t}$:

$$y_{m,t} = s_t e^{i\theta_{m,t}} + \epsilon_{m,t}, \qquad m = 0, \ldots, M-1. \tag{2}$$

Only relative offsets are observable. The transformation

$$\begin{aligned} \psi_t &\mapsto \psi_t + \gamma_t, \\ \theta_{m,t} &\mapsto \theta_{m,t} - \gamma_t \end{aligned} \tag{3}$$

leaves every noiseless observation unchanged. A global phase therefore cannot be recovered from these observations alone. The benchmark fixes the reference gauge by setting

$$\theta_{0,t} = 0. \tag{4}$$

This is a coordinate choice, not proof that module zero has privileged physical status. A deployment could instead use a graph-wide angular-synchronization estimator, an external clock, or a learned gauge tied to a task target.

The model also contains a more serious identifiability requirement. If module m does not observe the same latent process, then the product of its state with the reference state does not isolate a meaningful relative offset. No rotation estimator can repair the absence of a shared variable merely by increasing coherence.

5. A Causal Relative-Phase Estimator

For each module, form the normalized cross-product with the reference:

$$r_{m,t} = \frac{y_{m,t} y_{0,t}^*}{\max\left(\left|y_{m,t} y_{0,t}^*\right|, \varepsilon\right)}. \tag{5}$$

Here $\operatorname{unit}(z) = z / \max(|z|, \varepsilon)$, using the same numerical stabilizer as Equation (5).

In the low-noise shared-state case, the common latent phase cancels and $r_{m,t}$ approaches $e^{i\theta_{m,t}}$. The estimator is initialized from the entire calibration mean and then updated from the previous observation:

$$\begin{aligned} q_m^{\text{cal}} &= \operatorname{unit}\left[\frac{1}{T_{\text{cal}}} \sum_{u=0}^{T_{\text{cal}}-1} r_{m,u}\right], \\ q_{m,t} &= \begin{cases} q_m^{\text{cal}}, & t < T_{\text{cal}}, \\ \operatorname{unit}\left[(1-\eta) q_{m,t-1} + \eta r_{m,t-1}\right], & t \ge T_{\text{cal}}, \end{cases} \qquad 0 < \eta \le 1, \end{aligned} \tag{6}$$

with $q_0^{\text{cal}} = q_{0,t} = 1$ in the declared reference gauge. The one-step delay after calibration prevents the estimator from using the current target state to align itself. The phase estimate is

$$\widehat{\theta}_{m,t} = \arg q_{m,t}. \tag{7}$$

Each state is rotated into the reference gauge:

$$\widetilde{y}_{m,t} = y_{m,t} e^{-i\widehat{\theta}_{m,t}}. \tag{8}$$

The candidate fused estimate is

$$\widehat{s}_t^{\text{adapt}} = \frac{1}{M} \sum_{m=0}^{M-1} \widetilde{y}_{m,t}. \tag{9}$$

A static control estimates one offset from the calibration interval and never updates it:

$$\widehat{\theta}_m^{\text{static}} = \arg\left[\frac{1}{T_{\text{cal}}} \sum_{t<T_{\text{cal}}} r_{m,t}\right]. \tag{10}$$

The reference-only baseline is

$$\widehat{s}_t^{\mathrm{ref}} = y_{0,t}. \tag{11}$$

The benchmark also includes an unaligned complex mean, a phase-blind scalar reliability gate, static alignment, and an oracle that uses the known synthetic offsets of modules that truly observe the shared state.

6. Task Metric and Validation Gate

For an interval $\mathcal{I}$, normalized mean-squared error is

$$\mathrm{NMSE}_{\mathcal{I}}(\widehat{s}) = \frac{\sum_{t\in\mathcal{I}} |\widehat{s}_t - s_t|^2}{\sum_{t\in\mathcal{I}} |s_t|^2}. \tag{12}$$

The gain η is selected only on the validation interval:

$$\eta^* = \arg\min_{\eta\in\mathcal{G}} \ \mathrm{NMSE}_{\mathrm{val}}\left(\widehat{s}_\eta^{\mathrm{adapt}}\right), \tag{13}$$

where

$$\mathcal{G} = \{0.005, 0.01, 0.02, 0.04, 0.08, 0.16, 0.30\}. \tag{14}$$

Adaptive fusion is accepted only if it improves validation NMSE by at least ten percent relative to the reference:

$$A = \nVdash \left[\mathrm{NMSE}_{\mathrm{val}}\left(\widehat{s}_{\eta^*}^{\mathrm{adapt}}\right) \leq 0.90 \ \mathrm{NMSE}_{\mathrm{val}}\left(\widehat{s}^{\mathrm{ref}}\right)\right]. \tag{15}$$

The selected output, abbreviated sel in Equation (16), is

$$\widehat{s}_t^{\mathrm{sel}} = \begin{cases} \widehat{s}_t^{\mathrm{adapt}}, & A = 1, \\ \widehat{s}_t^{\mathrm{ref}}, & A = 0. \end{cases} \tag{16}$$

Equation (16) is the principal safety and scientific honesty feature of this revision. The proposed mechanism is not entitled to replace the baseline when its held-out evidence is inadequate.

Two descriptive diagnostics accompany task error. Let $\mathcal{M}_v$ be the set of valid nonreference modules: modules $m \neq 0$ that observe the shared latent. Circular offset error is

$$E_\theta = \frac{1}{|\mathcal{I}||\mathcal{M}_v|} \sum_{t\in\mathcal{I}} \sum_{m\in\mathcal{M}_v} \left|\mathrm{wrap}\left(\widehat{\theta}_{m,t} - \theta_{m,t}\right)\right|, \tag{17}$$

and normalized population coherence is

$$C = \frac{1}{|\mathcal{I}|} \sum_{t\in\mathcal{I}} \frac{|\sum_m \widetilde{y}_{m,t}|}{\max(\sum_m |\widetilde{y}_{m,t}|, \varepsilon)}. \tag{18}$$

High C is not accepted as success by itself. Distractor modules can be rotated into high apparent agreement even when the fused task estimate becomes worse. Task error and held-out acceptance remain governing.

7. Executable Benchmark

7.1 Frozen configuration

Each run contains 2,048 time steps and eight modules. The first 256 steps initialize calibration, the next 384 steps select η and decide acceptance, and the remaining 1,408 steps form the untouched test interval. Complex Gaussian observation noise has standard deviation 0.125. Each scenario uses 48 deterministic seeds.

The latent amplitude and phase both change through time. Module zero remains in the reference gauge. The other modules follow one of five offset processes.

7.2 Five worlds

World	Data-generating condition	Intended question
Stable offsets	Every module observes the shared latent state through one fixed offset.	Can calibration recover a reusable alignment?
Slow drift with jump	Offsets follow a slow random walk and undergo one moderate discontinuity.	Can a causal estimator track recoverable nonstationarity?
Fast drift	Offsets follow a random walk with 0.40-radian innovation scale.	Does estimator bandwidth become inadequate?
Unpredictable offsets	Every nonreference offset is independently redrawn at each step.	Does the gate reject a relation that past data cannot predict?
Distractor modules	Four modules observe the shared state; four follow independent complex trajectories.	Does apparent alignment fail when the common-latent assumption is false?

7.3 Baselines

Method	Role
Reference module	Safe single-observer fallback; no multisource variance reduction.
Naive complex mean	Tests premature fusion without offset correction.
Scalar reliability gate	Tests whether phase-blind calibration weights explain the result.
Static phase alignment	Tests whether one calibration is sufficient.
Adaptive phase alignment	Tests the proposed causal offset tracker.
Validation-gated selection	Tests whether held-out abstention prevents replacement by a failing estimator.
Oracle valid-module alignment	Uses true synthetic offsets and excludes known distractors; an unattainable ceiling, not a deployable method.

7.4 Reproducibility

The executable source is:

model/run_phase_alignment_benchmark.py

The contract tests are:

model/test_phase_alignment_benchmark.py

The frozen output is:

model/results-v1.json

The output contains every seed-level result, the full gain grid, scenario summaries, configuration, and a content hash. No stochastic value is inferred from the manuscript table.

The real-signal extension was frozen before confirmatory execution in PREREGISTRATION-REAL-SIGNAL-ECG-V2-20260728.md. Its executable, contract tests, and frozen output are:

- model/run_real_signal_ecg_benchmark.py
- model/test_real_signal_ecg_benchmark.py
- model/real-signal-ecg-results-v2.json

The v1 preregistration is retained as the one-seed runtime-pilot record. The v2 record freezes resampling, splits, gain grid, 48 seeds, baselines, failure conditions, bootstrap interval, Wilcoxon tests, and Holm familywise correction before the complete five-world run.

8. Results

Table 1 reports median NMSE on the held-out synthetic test interval. Acceptance is the fraction of 48 runs in which adaptive alignment passed Equation (15).

World	Reference	Naive mean	Scalar gate	Static alignment	Adaptive alignment	Selected output	Oracle	Acceptance
Stable offsets	0.014789	0.921575	0.903932	0.001879	0.001865	**0.001865**	0.001849	48/48
Slow drift with jump	0.014854	0.931799	1.077596	0.035235	0.002591	**0.002591**	0.001854	48/48
Fast drift	**0.014855**	0.888972	0.886402	0.896441	0.051210	**0.014855**	0.001848	0/48
Unpredictable offsets	**0.014772**	0.876694	0.874074	0.874044	0.877757	**0.014772**	0.001852	0/48
Distractor modules	**0.014789**	0.854633	0.889722	0.312904	0.025032	**0.014789**	0.003686	0/48

In the stable world, selected alignment reduced median NMSE by 87.4 percent relative to the reference. In the slow-drift world, it reduced median NMSE by 82.6 percent. These improvements approach but do not equal the oracle ceiling.

The static estimator succeeded under stable offsets but failed after drift and a jump. The adaptive estimator tracked slow change: its median test circular offset error was 0.0394 radians, compared with 0.4063 radians for the static estimate. Under fast drift, adaptive error rose to 0.4594 radians and adaptive NMSE exceeded the reference; the gate correctly abstained.

The distractor case is particularly informative. Adaptive alignment produced a median coherence of 0.971 while its NMSE was worse than the reference. High internal agreement therefore did not demonstrate correct reconstruction. Equation (15), rather than coherence alone, prevented promotion.

8.1 Preregistered semi-synthetic ECG result

The second benchmark tests whether the result depends on a generated sinusoidal latent. The source is the five-minute, 360 Hz ECG excerpt from record 208 of the MIT-BIH Arrhythmia Database distributed by SciPy [13,14]. Its stored SHA-256 is F20AD3365FB9B7F845D0E5C48B6FE67081377EE466C3A220B7F69F35C8958BAF. The complete recording was converted to millivolts, standardized, resampled from 360 Hz to 36 Hz with a polyphase anti-aliasing filter, and transformed into an analytic signal. The data were not selected by rhythm, event, or diagnosis. Controlled phase offsets, observation noise, and distractor trajectories were then applied to that fixed real waveform.

Table 2 reports median test NMSE over 48 perturbation seeds per world. The paired interval is the preregistered deterministic 95 percent bootstrap interval for the median seed-level difference `selected minus reference`.

World	Reference	Static	Adaptive	Selected	Oracle	Acceptance	Paired difference 95% CI
Stable offsets	0.008947	0.001130	0.001173	**0.001173**	0.001121	48/48	[-0.007804, -0.007732]
Slow drift with jump	0.008956	0.015817	0.001327	**0.001327**	0.001119	48/48	[-0.007637, -0.007467]
Fast drift	0.008955	0.529374	0.002356	**0.002356**	0.001121	48/48	[-0.006642, -0.006521]
Unpredictable offsets	**0.008977**	0.875469	0.875488	**0.008977**	0.001119	0/48	[0, 0]
Distractor modules	**0.008947**	0.323366	0.238344	**0.008947**	0.002237	0/48	[0, 0]

The stable, slow-drift, and fast-drift paired differences remained significant after Holm correction across the five planned comparisons (adjusted `p = 3.55e-14` for each). The two abstention worlds returned the reference exactly, so their paired differences were all zero and their adjusted `p` values were 1. These probabilities quantify repeated controlled perturbations of one record; they do not count 48 patients or 48 independent biological samples.

The accepted fast-drift result does not contradict the synthetic fast-drift failure. The two declarations use different sampling scales and innovation magnitudes. Together they locate rather than erase the estimator's bandwidth boundary: 'fast' is not an invariant label, and performance must be reported relative to sample rate, offset innovation, gain grid, signal-to-noise ratio, and validation rule.

9. What the Benchmark Establishes

The result supports four bounded propositions inside the declared synthetic model:

1. Relative cyclic offsets can be estimated causally from normalized cross-products when modules observe one shared complex state.
2. Static calibration is adequate for fixed offsets but not for the declared drift-and-jump process.
3. A causal estimator has a finite tracking bandwidth; rapid or unpredictable offsets defeat it.
4. A held-out task gate can prevent a failed alignment model from replacing a safer baseline in the tested worlds.

The semi-synthetic extension adds two bounded propositions:

5. The improvement is not restricted to the generated sinusoidal latent; it recurs when the latent trajectory is derived from one complete real ECG waveform under controlled offsets and noise.
6. Estimator bandwidth is relative to the sampling and perturbation scale; therefore a categorical label such as fast drift is insufficient without numerical specification.

The results do not show that complex-valued fusion is universally superior. They show the opposite: phase alignment is conditional, and the conditions must be tested.

10. Multiscale Aperture Synthesis Imaging as the Engineering Source

The motivating engineering source is Wang et al.'s multiscale aperture synthesis imager (MASI) [1]. MASI addresses distributed coherent imaging with coded lensless sensors. It belongs to a broader computational-imaging lineage that includes Fourier ptychographic synthesis of a wider numerical aperture from multiple measurements [3]. The relevant MASI source mechanism is:

1. distributed coded sensors capture local diffraction information;
2. each sensor recovers a complex wavefield;
3. reconstructed wavefields are digitally padded;
4. each wavefield is propagated to the object plane;
5. relative sensor phase offsets are optimized;
6. phase-corrected fields are fused in real space; and
7. overlapping measurement regions are not required for the reported alignment procedure.

The engineering chain is:

```
local coded capture
-> preserve complex wavefield
-> solve relative offsets
-> phase-correct
-> coherent reconstruction
```

MASI does not implement Equations (5)-(16), and the synthetic benchmark is not a reproduction of MASI. MASI supplies a concrete demonstration that alignment-relevant information can be retained locally and used to reconstruct coherence computationally after capture. The benchmark tests a generalized module-fusion analogue with explicit abstention.

11. Translation Across Domains

The word *phase* changes meaning across domains. The mappings below are functional comparisons, not identities.

Term	Optics / MASI	Biological research	AI design in this paper
Local observer	Coded optical sensor	Neuron, ensemble, column, or measured population	Module emitting a structured state
Phase	Optical complex-wavefield phase	Relative timing in oscillatory electrical activity	Circular coordinate of a complex state
Offset estimator	Reconstruction-time phase optimization	Candidate timing, coupling, inhibitory, or predictive process	Causal normalized cross-product tracker
Alignment	Coherent object-plane fusion	Task-dependent coordination or phase relation	Rotation into one declared gauge
Gate	Reconstruction objective and experimental conditions	Attention, inhibition, state, and behavior	Held-out NMSE acceptance rule
Failure	Defocus or incoherent reconstruction	Loss of predictive or behavioral relation after controls	Worse task error, unpredictable offset, or no shared latent

Three category errors remain prohibited:

1. optical phase is not automatically neural phase;
2. a digital optical optimizer is not automatically a cortical mechanism; and
3. a high coherence score is not evidence of consciousness.

12. SAN, NAPOT, and Phase Wave Differentials

Self Aware Networks enters only after the engineering operation has been specified. SAN proposes that receiving neural structures are already in a changing context; the consequence of an arriving event depends on the receiver, its prior state, local inhibition, route, and relative timing. **Phase Wave Differential** is SAN's name for a candidate receiver-relative difference that can carry consequential update information. **Neural Array Projection Oscillation Tomography** names a larger proposed receive-transform-project cycle by which distributed arrays repeatedly update partial renderings.

Draft 6 also preserved a broader SAN source vocabulary: gamma oscillations, thalamic gating, alpha / beta inhibition, synchronized bursts, BOT, COT, coincidence bits, inhibitory-interneuron gating, oscillatory integration, tomographic projection, and frequency maps. Those labels remain part of the paper's source genealogy; this synthetic benchmark does not validate them. Within that historical model vocabulary, a coincidence bit treats near-concurrent arrivals as a candidate relational event, while COT and NAPOT name proposed cellular and neural-array reconstruction scales. They are kept visible here so a technical improvement does not silently erase the biological research program that motivated Draft 6.

This is not a renaming of angular synchronization, phase coding, communication-through-coherence, predictive coding, traveling waves, or complex-valued neural networks. Those inherited concepts address related but different questions:

Existing concept	What it already addresses	Remaining SAN / PWD question
Angular synchronization	Recovery of unknown angles from relative offsets	What biological receiver-relative event is being estimated, routed, and used?
Phase coding	Information associated with spike or oscillation phase	How does the receiver's current state transform an arriving relation into a downstream difference?
Communication through coherence	Phase relations as a mechanism of effective communication [4]	What exact local difference triggers rerouting, plasticity, or reconstruction?
Predictive coding	Error between predictions and incoming signals [5]	How are timing, receiver state, and distributed projection joined in the proposed SAN operator?
Traveling waves	Propagating spatiotemporal neural activity	What is the receive-transform-project operation at each downstream array?
Complex-valued AI	Computation with magnitude and phase-like coordinates	When does explicit causal offset solving beat compute-matched alternatives and when must it abstain?

The distinction matters because a new term is warranted only if it names an operation the older terms do not already specify. In this paper, PWD is not used as a synonym for the estimator. It is a biological hypothesis that would require its own sender, receiver, delay, expected relation, perturbation, and downstream consequence to be measured.

13. Empirical AI Program

The synthetic result earns a next experiment; it does not substitute for one.

Task	Minimal perturbation	Primary comparison	Failure condition
Asynchronous audio-video fusion	Variable delay and clock drift	Cross-attention, scalar gating, temporal convolution	No improvement in held-out classification or alignment error
Camera-IMU fusion	Drift and discontinuous reset	State-space or Kalman-style alignment with equal compute	No reduction in pose or event-prediction error
Multi-agent state reconciliation	Delayed, missing, and incompatible observations	Message-passing and attention baselines	Apparent coherence rises while state error does not fall

Task	Minimal perturbation	Primary comparison	Failure condition
Identity persistence	Occlusion and noisy reappearance	Recurrent and contrastive trackers	False matches rise or identity continuity does not improve
Memory-context drift	Stale or shifted memory state	Recurrent memory and retrieval baselines	Fused state increases inconsistency or hallucinated updates

The complete compute-matched comparator suite inherited from Draft 6 is:

Comparison class	Required baseline
Scalar fusion	Weighted averaging, learned scalar gates, and mixture-of-experts routing
Attention	Transformer self-attention and cross-attention [6]
Recurrent memory	RNN, LSTM, GRU, or a state-space model with equivalent parameter and compute budgets
Predictive alignment	Predictive-coding or error-correction model without a complex or cyclic phase coordinate
Contrastive fusion	CLIP-style or other multimodal contrastive objective [12]
Complex-valued control	Complex-valued neural network with no explicit offset-solving pass

Required ablations include removing the circular head, randomizing offsets, freezing offsets, removing the held-out gate, replacing phase with scalar confidence, varying reference choice, changing estimator bandwidth, and compute-matching every baseline.

A real AI claim requires preregistered tasks, independent runs, uncertainty intervals, multiplicity control, parameter and compute accounting, and a sealed evaluation set not used to design the acceptance rule.

14. Neuroscience Program

The biological mapping must also be able to lose.

Candidate proposition	Measurement	Required controls	Failure condition
Relative timing improves task integration	Phase relation, firing rate, task performance	Stimulus energy, common drive, volume conduction, rate-only model	Timing adds no out-of-sample explanatory value
Inhibitory timing resets or redirects a coalition	Cell-type-specific timing perturbation; measure phase-dispersion contraction, order-parameter change, and behavior	Nonspecific inhibition, arousal, movement, heating	No contraction, order-parameter change, or task-relevant behavioral effect
Receiver-relative mismatch predicts update	Sender, receiver, delay, expected relation, downstream response	Prediction error without phase, random-phase null, history effects	Mismatch does not predict correction after controls
Recurrent arrays update a distributed scene estimate	Multisite measurement plus perturbation and behavioral report	Feedforward, recurrent, predictive, and attention alternatives	No discriminating projection / reconstruction signature
SAN / NAPOT labels improve evidence organization	Use Paper 09 graph labels to separate mainstream anchor, model term, conjecture, and rejected edge	Same evidence classified without SAN labels	Labels do not improve classification or experiment design

These tests treat communication-through-coherence and predictive processing as serious comparators. A SAN label is useful only if it specifies a measurable operation and a result that distinguishes it from those inherited accounts.

15. Intent Recovery and Lossless Lineage

Draft 6 remains unchanged and is pinned in the internal release validation by SHA-256:

5B36D0024DC3BB1650E90CA6955296B147C565A84CEDCA4E875ABCC406970932

Draft 7 preserves its substantive architecture atom by atom:

Draft 6 surface	Draft 7 preservation
MASI engineering source	The seven-step engineering source and local-capture-to-coherent-reconstruction chain remain in Section 10.
Domain boundary	The optics / biology/AI translation boundary and prohibition on promoting analogy into validation remain in Section 11.
SAN / NAPOT vocabulary	Gamma oscillations, thalamic gating, alpha / beta inhibition, synchronized bursts, BOT, COT, NAPOT, coincidence bits, PWD, inhibitory-interneuron gating, oscillatory integration, tomographic projection, and frequency maps remain in Section 12.
AI architecture	Module state, phase head, offset estimator, coherence objective, task head, benchmark tasks, full baseline family, and ablations remain in Sections 3-7 and 13.
Neuroscience tests	Phase integration, inhibitory reset, phase-dispersion contraction, order-parameter change, receiver-relative mismatch, and Paper 09 label tests remain in Section 14.
SAN Scaling route	The private `SAN Scaling.txt` route, including its historical `sqrt(N)` to `N`, reset amplification, and order-parameter bridge, remains in the source ledger; this benchmark does not establish that scaling claim.
Cross-paper controls	Paper 09 claim labels, Paper 14 translation boundaries, and Paper 16 provenance controls remain explicit source-ledger dependencies.
Figures and sources	The four-lane transfer figure plan and private / public source ledger remain in Sections 16-17.
Draft 7 executable result	Every synthetic world, seed count, equation, result table, negative control, and claim boundary remains in Sections 4-9; the ECG extension is additive.

Earlier private source language is treated as an intent-recovery record. A compressed or overstrong statement is not silently deleted and is not copied unchanged into current science. The original wording and date remain in its source; the current manuscript recovers the strongest defensible operation, names inherited prior art, and states the remaining hypothesis separately. Nothing developed in 2026 is assigned an earlier public date.

16. Source and Reproducibility Ledger

Source	Location	Use	Boundary
Published Draft 6	Predecessor manuscript; SHA-256 shown in Section 15	Complete predecessor architecture	Preserved, not overwritten

Source	Location	Use	Boundary
Published PDF	Existing Computational Phase Synchronization record	Public-version byte baseline	Not modified by this package
MASI primary paper	Wang et al. 2025; DOI in Reference 1	Optical mechanism and equations	Engineering source, not biology
MASI source-development record	Private author archive	Private development route	Provenance, not public scholarship
CPS source manuscript	Private author archive	Historical architecture and terminology	Private source
CPS companion PDF	Private author archive	Historical paper artifact	Private source, not an external citation
AI Studio CPS PDF	Private author archive	Export and source route	Private export
April 2024 SAN / NAPOT development record	Private author archive	Private terminology-development route	Does not move a public date
SAN OCA Draft 1	Private author archive	Gamma, thalamus, NAPOT, and PWD intent recovery	Source genealogy, not validation
Building Sentient Beings	Authored companion paper and private source archive	Agent / module / world-model architecture	Companion architecture, not validation
SAN Scaling source	Private author archive	Historical `sqrt(N)` to `N`, reset amplification, and order-parameter bridge	Private model route; not validated here
Paper 09 Draft 6	Companion paper package	Mainstream-anchor, model-term, conjecture, and rejected-edge labels	Internal cross-paper control
Paper 14 Draft 6	Companion paper package	Translation-boundary and demotion vocabulary	Internal cross-paper control
Paper 16 Draft 6	Companion paper package	Public / private source handling and provenance-is-not-proof rule	Internal cross-paper control
Benchmark source	model/run_phase_alignment_benchmark.py	Frozen data-generating processes and estimators	Synthetic only
Benchmark tests	model/test_phase_alignment_benchmark.py	Executable acceptance contract	Tests code, not external reality
Frozen results	model/results-v1.json	Seed-level and aggregate results	48 seeds per declared world
ECG preregistration v1	PREREGISTRATION-REAL-SIGNAL-ECG-V1-20260728.md	One-seed timing-pilot record	Superseded before confirmatory execution
ECG preregistration v2	PREREGISTRATION-REAL-SIGNAL-ECG-V2-20260728.md	Frozen real-signal hypotheses, splits, uncertainty, multiplicity, and failure rules	Confirmatory authority
ECG source	data-cache/ecg.dat	Checksum-verified SciPy copy of MIT-BIH record 208 excerpt	Real waveform; no medical labels used
ECG benchmark source and tests	model/run_real_signal_ecg_benchmark.py; model/test_real_signal_ecg_benchmark.py	Semi-synthetic execution and contract tests	Not a clinical or general-AI test
Frozen ECG result	model/real-signal-ecg-results-v2.json	Seed-level results, intervals, and multiplicity correction	One source record with controlled perturbations

17. Figure Plan

The principal publication figure should retain Draft 6's four-lane transfer diagram while adding the validation fork:

```
MASI optics
local coded capture -> relative offset solving -> coherent reconstruction

General module architecture
local complex states -> causal relative-phase estimate -> aligned candidate

Validation decision
held-out task improves? -> yes: fuse
                        -> no: reference fallback

Biological research
sender / receiver timing -> candidate receiver-relative difference -> perturb and test
```

Caption: *Computational phase synchronization as a conditional transfer pattern. MASI supplies an established optical engineering source. The synthetic module lane implements a causal reference-gauge estimator. Held-out validation decides whether fusion is permitted. The biological lane remains a testable translation rather than a validated mechanism.*

18. Limitations

The benchmark is deliberately simple. It uses one complex latent scalar, one reference module, Gaussian observation noise, a fixed gain grid, equal fusion weights, known evaluation targets, and hand-declared scenario families. The acceptance threshold was not independently preregistered. Forty-eight seeds are replications of synthetic random worlds, not independent real datasets.

The reference module can fail, the shared latent can be multidimensional, phase can be locally ambiguous, delays can interact with content, pairwise graphs can be sparse, module reliability can vary, and an adversary can overfit the validation gate. Future work should compare graph-wide angular synchronization, state-space delay estimation, Bayesian uncertainty, robust module rejection, and independently sealed task suites.

The real-signal extension uses one ECG record, resampled from 360 Hz to 36 Hz. Its 48 seeds are repeated phase / noise perturbations of that one trajectory, not independent people or recordings. Module offsets and distractors remain simulated, and the known latent makes reconstruction error directly observable. The result therefore tests waveform robustness of the estimator, not clinical generalization, event recognition, sensor synchronization in the wild, or neural communication.

Most importantly, successful reconstruction does not imply consciousness. Coherence is neither necessary nor sufficient for a conscious state in this paper. The contribution is a conditional fusion mechanism with an explicit failure route.

19. Conclusion

Distributed states should not be fused merely because they can be made to look coherent. They should retain alignment-relevant structure, expose the relative offset problem, and earn fusion through held-out task performance.

The synthetic benchmark shows both sides of that principle. Causal adaptive alignment sharply improved reconstruction under stable and slowly drifting relative phases. It failed under its declared fast drift, unpredictable offsets, and unrelated distractor modules; the validation gate preserved the reference in all three negative worlds.

The preregistered ECG extension reproduced the improvement for stable and slow-drift offsets and also succeeded at its numerically milder, resampled fast-drift condition. It rejected all unpredictable-offset and distractor runs. This comparison makes the bandwidth claim more precise: recoverability depends on the ratio among offset dynamics, sampling interval, estimator gain, and signal quality.

The defensible result is therefore conditional: computational phase synchronization can be useful when a shared state exists and relative offsets are predictable at the estimator's bandwidth. The same architecture must abstain when those assumptions do not hold. That conditional mechanism now provides a concrete starting point for real AI benchmarks and a carefully bounded comparator for SAN, NAPOT, and PWD experiments.

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